

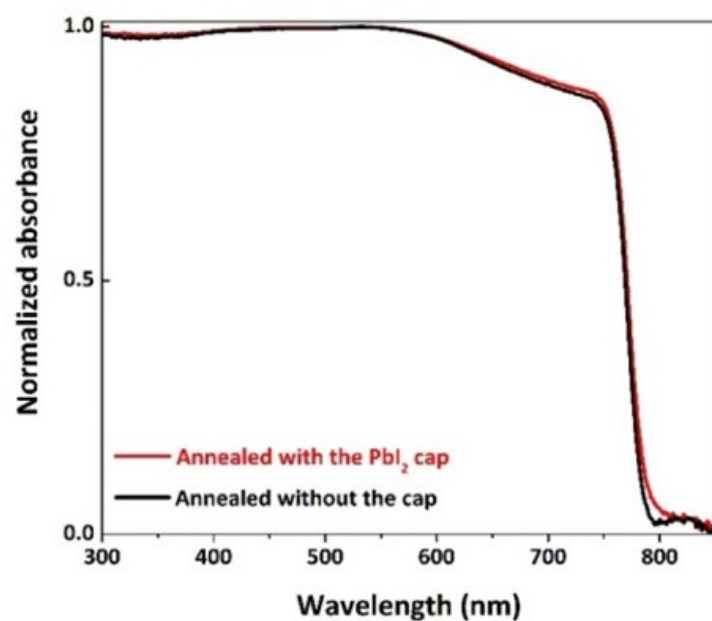


Fig. S10 Normalized UV-Vis spectra of the $\text{CH}_3\text{NH}_3\text{PbI}_3$ perovskite films annealed with and without the PbI_2 cap.

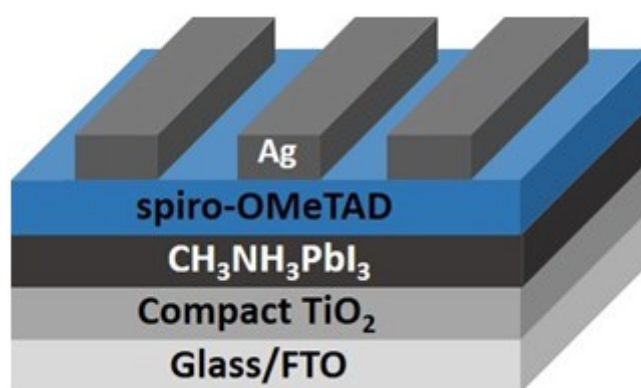


Fig. S11 Schematic of the structure of the fabricated perovskite solar cell.

Table S1 Summary of the PV performance parameters of the fabricated perovskite solar cell with the $\text{CH}_3\text{NH}_3\text{PbI}_3$ perovskite film annealed with the PbI_2 cap, TiO_2 cap, or without cap.

Sample	$J_{sc}(\text{mA}/\text{cm}^2)$	$V_{oc}(\text{V})$	FF	PCE(%)
With PbI_2 cap	23.41	1.07	0.70	17.57
Without cap	22.62	1.05	0.60	14.19
With TiO_2 cap	19.28	0.99	0.66	12.62